

LAYOUT SCRIPT INTERFACE

View online: <https://www.construct.net/en/make-games/manuals/construct-3/scripting/scripting-reference/layout-interfaces/ilayout>

The `ILayout` script interface represents a layout in the project.

Getting an ILayout

The `ILayout` interface is typically accessed via the `IRuntime` `layout` property, e.g. `runtime.layout`. This represents the current running layout. Other layouts can be accessed via the `IRuntime` methods `getLayout()` and `getAllLayouts()`.

Layout events

The following events can be listened for using the `addEventListener` method.

"beforelayoutstart"

"afterlayoutstart"

Fired when the layout starts. `"beforelayoutstart"` fires just before *On start of layout*, and `"afterlayoutstart"` fires just after. In both events, all instances on the layout are created and available to modify.

These events can use async handler functions, and the runtime will wait for them to finish before continuing.

"beforelayoutend"

"afterlayoutend"

Fired when the layout ends due to changing to another layout. `"beforelayoutend"` fires just before *On end of layout*, and `"afterlayoutend"` fires just after. In both events, all instances on the layout are still available, but all non-global instances are destroyed immediately after the `"afterlayoutend"` event.

These events can use async handler functions, and the runtime will wait for them to finish before continuing.

Layout APIs

runtime

A reference back to the `IRuntime` interface.

name

A read-only string of the layout name.

index

A read-only number of the zero-based index of the layout in the order it appears in the Project Bar.

addEventListener(eventName, callback)

removeEventListener(eventName, callback)

Add or remove a callback function for an event. See *Layout events* above for the available events.

width

height

setSize(width, height)

getSize()

Set or get the size of the layout. The methods allow setting and getting both values at the same time.

Note a layout cannot have a zero or negative size.

scrollX

scrollY

scrollTo(x, y)

getScrollPosition()

Set or get the scroll position in layout co-ordinates. `scrollTo()` is a shorthand for setting both `scrollX` and `scrollY`, and `getScrollPosition()` returns both scroll co-ordinates at the same time.

isUnboundedScrolling

A read-only boolean reflecting the state of the *Unbounded scrolling* layout property.

sampling

A string of the layout's sampling mode, which is one of `"auto"`, `"nearest"`, `"bilinear"` or `"trilinear"`.

activeSampling

A read-only string with the layout's active sampling mode. This is only different to `sampling` when the sampling mode is `"auto"`: in this case the active sampling mode is inherited from the project setting.

scale

Set or get the layout scale, with `1` being the default scale, `2` being 2x scale, etc. This scales all the layers in the layout, taking in to account their scale rate property.

angle

Set the layout angle in radians. This rotates all the layers in the layout.

projection

Set or get a string specifying the current layout projection, which must be one of `"perspective"` or `"orthographic"`. For more details see *Projection* in [Layout Properties](#).

setVanishingPoint(vpX, vpY)

getVanishingPoint()

Set or get the *Vanishing point* [layout property](#), with each component in the range 0-1. The getter returns an array with two elements in the form `[vpX, vpY]`.

effects

An array of [IEffectInstance](#) representing the effect parameters of the effects on this layout.

Layer APIs

These APIs relate to the set of layers on the layout.

getLayer(layerNameOrIndex)

Get an [ILayer interface](#) for a layer on the layout, by a case-insensitive string of its name or its zero-based index. When passing a number, an out-of-range number is clamped to the valid range and the nearest layer returned. When passing a string, if no layer with the given name is found, the method returns `null`.

*allLayers()

Iterates **ILayer interfaces** representing all the layers on the layout, in increasing Z order.

getAllLayers()

Return an array of **ILayer interfaces** representing all the layers on the layout, in increasing Z order.

addLayer(layerName, insertBy, where)

Create a new layer and insert it to the layer tree at runtime (also known as a *dynamic layer*).

layerName is a string of the name to use for the added layer, which must be different to all existing layers already added, including other dynamic layers. **insertBy** is an **ILayer** of another layer to insert the new layer relative to. **where** specifies where to insert the new layer relative to the **insertBy** layer, which may be one of the following strings:

- **"above"** or **"below"** : insert adjacent to the **insertBy** layer, above or below it in Z order.
 - **"top-sublayer"** or **"bottom-sublayer"** : insert as a sub-layer of the **insertBy** , at the top or bottom of its existing sub-layers. **insertBy** may also be **null** , in which case the new layer is added at the top or bottom at the root level of the layer tree.
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moveLayer(layerToMove, insertBy, where)

Remove and re-insert a layer to a new location in the layer tree. This works similarly to **addLayer()** , except **layerToMove** refers to an **ILayer** that already exists; otherwise the **insertBy** and **where** parameters are used in the same way.

removeLayer(layer)

Remove a given **ILayer** from the layer tree. This also removes any sub-layers of the removed layer, and all objects on the layer or any of its sub-layers will be destroyed. A layout must have at least one layer, so the last top-level layer cannot be removed.

removeAllDynamicLayers()

Removes all layers added using the **addLayer()** method, leaving only the layers added in the editor. All objects on the removed layers will be destroyed. This can be useful to reset the state of dynamic layers.